

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Know

Agentic Knowledgeable Self-Awareness for LLM-based Planning Systems

Core Idea

Enable agents to autonomously regulate cognitive mode by situation:

- Fast thinking: act directly when confident and correct
- Slow thinking: trigger self-reflection when initial action fails
- Knowledgeable thinking: consult external knowledge only when needed

Current Training is 'Flood Irrigation': Indiscriminate Knowledge Inject

Trajectory Imitation

- Uses gold trajectories indiscriminately
- No situational judgment
- Overfits to injected patterns

Trial-and-Error

- Relies on external feedback loops
- Costly iterative exploration
- Weak knowledge-use policy

Knowledge-Augmented

- Attaches knowledge to all states
- 100% retrieval usage common
- Efficiency and relevance mismatch

Human decision-making is situational; current agents mostly are not.

KnowSelf addresses this gap with agentic self-awareness over when to act, reflect, or retrieve.

Three Situational Modes and Decision Criteria

Current State

Fast Thinking

Directly output executable action
Criterion: immediate correctness

Slow Thinking

Trigger reflection / re-planning
Criterion: recoverable via internal reasoning

Knowledgeable Thinking

Query external knowledge, then plan
Criterion: requires missing world knowledge

KnowSelf Pipeline: Data Construction → Two-Stage Training → Self-A

1) Data Construction

- Self-explore trajectories in ALFWorld
- Heuristic situation judgment
- Mark special tokens: fast / slow / know

2) Two-Stage Training

- Stage A: Supervised fine-tuning (SFT)
- Stage B: Reasoning preference optimization (RPO)
- Learn when each mode should activate

3) Self-Aware Inference

- Model emits situational token
- Token routes behavior: act / reflect / retrieve
- Selective knowledge query only when needed

Framework reference: figure_3.jpg in source brief (pipeline overview)

Qualitative Examples: Reflection and Knowledge Retrieval in Practice

Task A: “put a hot mug in cabinet”

- KnowSelf: triggers slow thinking, heats mug, completes task
- OpenAI-O1 baseline: misses heat precondition, fails final placement

Task B: “put a saltshaker in drawer”

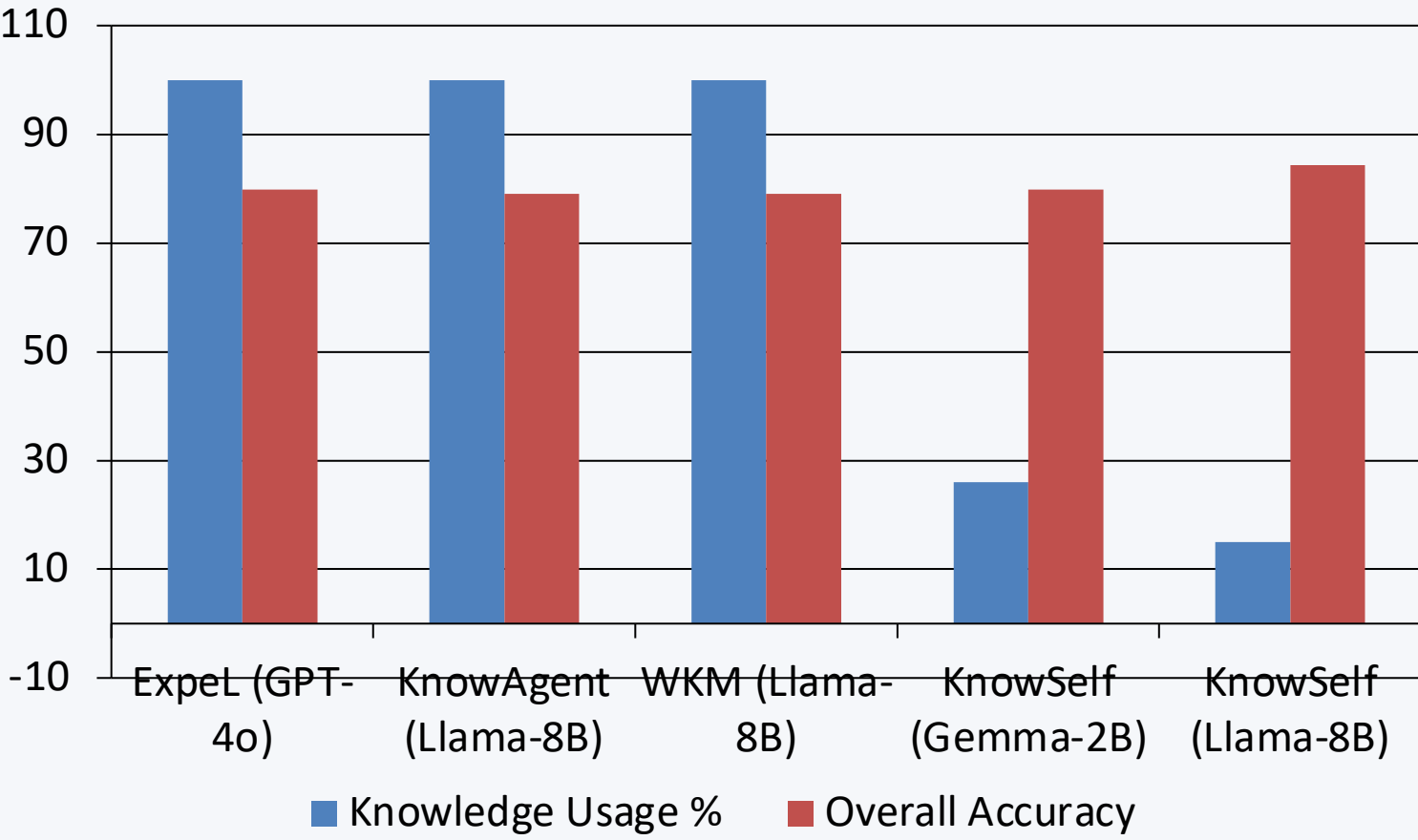
- KnowSelf: triggers knowledgeable thinking for placement rule
- DeepSeek-R1 baseline: loops without correct object-location strategy

Visual source reference: figure_1.jpg (side-by-side qualitative trajectories)

ALFWorld Results: KnowSelf Leads with Less Knowledge

Backbone	Method	Know%	Clean	Heat	Cool	Put	Examine	All
GPT-4o	ReAct	None	75.00	61.90	66.67	84.21	47.06	76.12
GPT-4o	Reflexion	None	75.00	66.67	66.67	78.95	47.06	76.12
GPT-4o	ExpeL	100%	75.00	66.67	76.19	78.95	52.94	79.85
Llama-8B	ReAct	None	71.88	57.14	61.90	73.68	47.06	71.64
Llama-8B	ETO	None	71.88	66.67	71.43	78.95	52.94	76.12
Llama-8B	KnowAgent	100%	68.75	57.14	80.95	84.21	52.94	79.10
Llama-8B	WKM	100%	65.63	57.14	80.95	84.21	52.94	79.10
Llama-8B	KnowSelf	15%	81.25	80.95	80.95	84.21	58.82	84.33
Gemma-2B	ReAct	None	65.63	57.14	57.14	73.68	41.18	70.15
Gemma-2B	KnowSelf	26%	65.63	73.91	76.19	83.33	52.94	79.85

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage



Key Efficiency Findings

- KnowSelf (Llama-8B): 84.33% at 15% knowledge
- KnowSelf (Gemma-2B): 79.85% at 26% knowledge
- Full-knowledge methods (100%) do not match top score
- Selective retrieval scales across 2B→8B backbones

Key Takeaways and Future Directions

Takeaways

- Indiscriminate knowledge injection is often wasteful and suboptimal in agent planning.
- Situational self-awareness can be learned via lightweight heuristic token labeling on self-explored trajectories.
- KnowSelf (Llama-8B) surpasses GPT-4o ReAct (84.33 vs 76.12) and GPT-4o ExpeL (84.33 vs 79.85).
- KnowSelf reaches SOTA-level outcomes with only 15–26% knowledge usage vs. 100% for several baselines.
- Future work: richer situation taxonomies, adaptive retrieval budgets, and transfer beyond ALFWorld.

KnowSelf framing: self-regulated cognition for efficient, robust LLM agents.